EDA

Load the Harmonized Data

```r
macro_df <- read_csv("../data/macro_panel_wide_raw.csv", show_col_types = FALSE) |>
  mutate(quarterly_date = as.Date(quarterly_date)) |>
  arrange(quarterly_date)

glimpse(macro_df)
```

```
## Rows: 197
## Columns: 9
## $ quarterly_date          <date> 1977-01-01, 1977-04-01, 1977-07-01, 1977-10-0~
## $ national_net_savings    <dbl> 15180, 14684, 13724, 14136, 13512, 14992, 1523~
## $ household_net_savings   <dbl> 13632, 17048, 13920, 15024, 19168, 17984, 1741~
## $ corporate_net_savings   <dbl> 4660, 3260, 3592, 4984, 3892, 3504, 4576, 3336~
## $ gross_domestic_income   <dbl> 37.1, 37.3, 37.4, 37.8, 38.0, 38.3, 38.6, 38.9~
## $ household_consumption   <dbl> 418612, 417567, 424081, 428064, 431165, 434052~
## $ unemployment_rate       <dbl> 7.8, 7.8, 8.3, 8.5, 8.5, 8.4, 8.4, 8.3, 7.9, 7~
## $ cpi_inflation_indicator <dbl> 32.8, 33.5, 34.2, 34.9, 35.7, 36.6, 37.1, 37.9~
## $ market_interest_rates   <dbl> 8.164062, 7.700000, 7.500000, 7.500000, 7.6307~
```

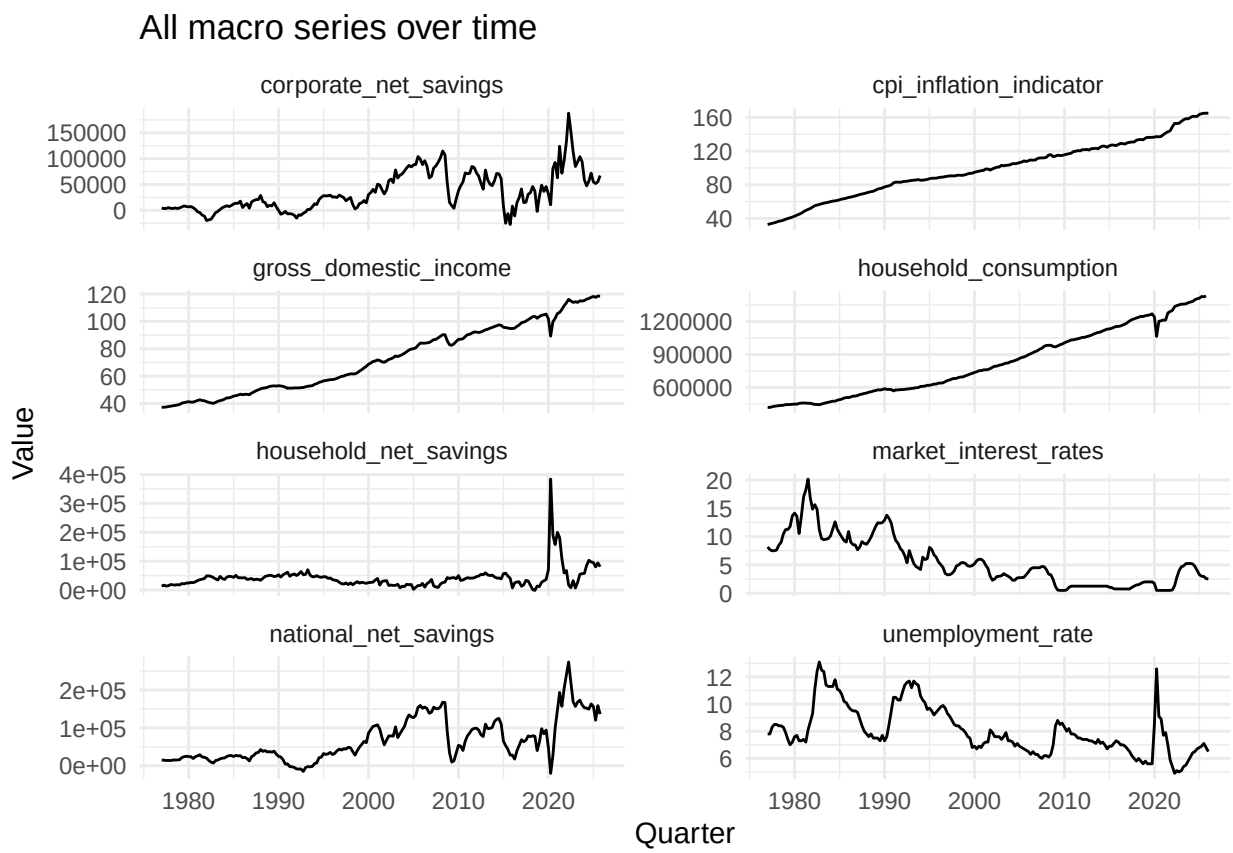
```r
summary(macro_df)
```

```
##  quarterly_date       national_net_savings household_net_savings
##  Min.   :1977-01-01   Min.   :-20116       Min.   : -1168
##  1st Qu.:1989-04-01   1st Qu.: 23627       1st Qu.: 23350
##  Median :2001-07-01   Median : 44448       Median : 37636
##  Mean   :2001-07-01   Mean   : 66616       Mean   : 41686
##  3rd Qu.:2013-10-01   3rd Qu.: 99984       3rd Qu.: 48448
##  Max.   :2026-01-01   Max.   :274012       Max.   :384240
##                       NA's   :1            NA's   :1
##  corporate_net_savings gross_domestic_income household_consumption
##  Min.   :-27516        Min.   : 37.10        Min.   : 417567
##  1st Qu.:  6463        1st Qu.: 51.38        1st Qu.: 570258
##  Median : 25036        Median : 71.10        Median : 761540
##  Mean   : 35195        Mean   : 72.68        Mean   : 826977
##  3rd Qu.: 61384        3rd Qu.: 94.42        3rd Qu.:1087645
##  Max.   :187928        Max.   :118.50        Max.   :1431258
##  NA's   :1             NA's   :1             NA's   :1
##  unemployment_rate cpi_inflation_indicator market_interest_rates
##  Min.   : 4.900    Min.   : 32.80          Min.   : 0.500
##  1st Qu.: 6.900    1st Qu.: 74.90          1st Qu.: 1.936
##  Median : 7.600    Median : 98.60          Median : 4.674
##  Mean   : 8.037    Mean   : 98.83          Mean   : 5.532
##  3rd Qu.: 9.100    3rd Qu.:123.30          3rd Qu.: 8.569
##  Max.   :13.100    Max.   :165.00          Max.   :20.153
##
```

```r
na_summary <- macro_df |>
  summarise(across(-quarterly_date, ~ sum(is.na(.)))) |>
  pivot_longer(
    cols = everything(),
    names_to = "series",
    values_to = "n_missing"
  ) |>
  arrange(desc(n_missing))

na_summary
```

```
## # A tibble: 8 x 2
##   series                  n_missing
##   <chr>                       <int>
## 1 national_net_savings            1
## 2 household_net_savings           1
## 3 corporate_net_savings           1
## 4 gross_domestic_income           1
## 5 household_consumption           1
## 6 unemployment_rate               0
## 7 cpi_inflation_indicator         0
## 8 market_interest_rates           0
```

```r
macro_long <- macro_df |>
  pivot_longer(
    cols = -quarterly_date,
    names_to = "series",
    values_to = "value"
  )

ggplot(macro_long, aes(x = quarterly_date, y = value)) +
  geom_line() +
  facet_wrap(~ series, scales = "free_y", ncol = 2) +
  theme_minimal() +
  labs(
    title = "All macro series over time",
    x = "Quarter",
    y = "Value"
  )
```




```r
eda_df <- macro_df |>
  select(
    quarterly_date,
    national_net_savings,
    household_net_savings,
    corporate_net_savings,
    gross_domestic_income,
    household_consumption,
    unemployment_rate,
    cpi_inflation_indicator,
    market_interest_rates
  ) |>
  rename(
    date = quarterly_date,
    nat_save = national_net_savings,
    hh_save = household_net_savings,
    corp_save = corporate_net_savings,
    gdi = gross_domestic_income,
    cons = household_consumption,
    unemp = unemployment_rate,
    cpi = cpi_inflation_indicator,
    rate = market_interest_rates
  ) |>
  arrange(date)

glimpse(eda_df)
```

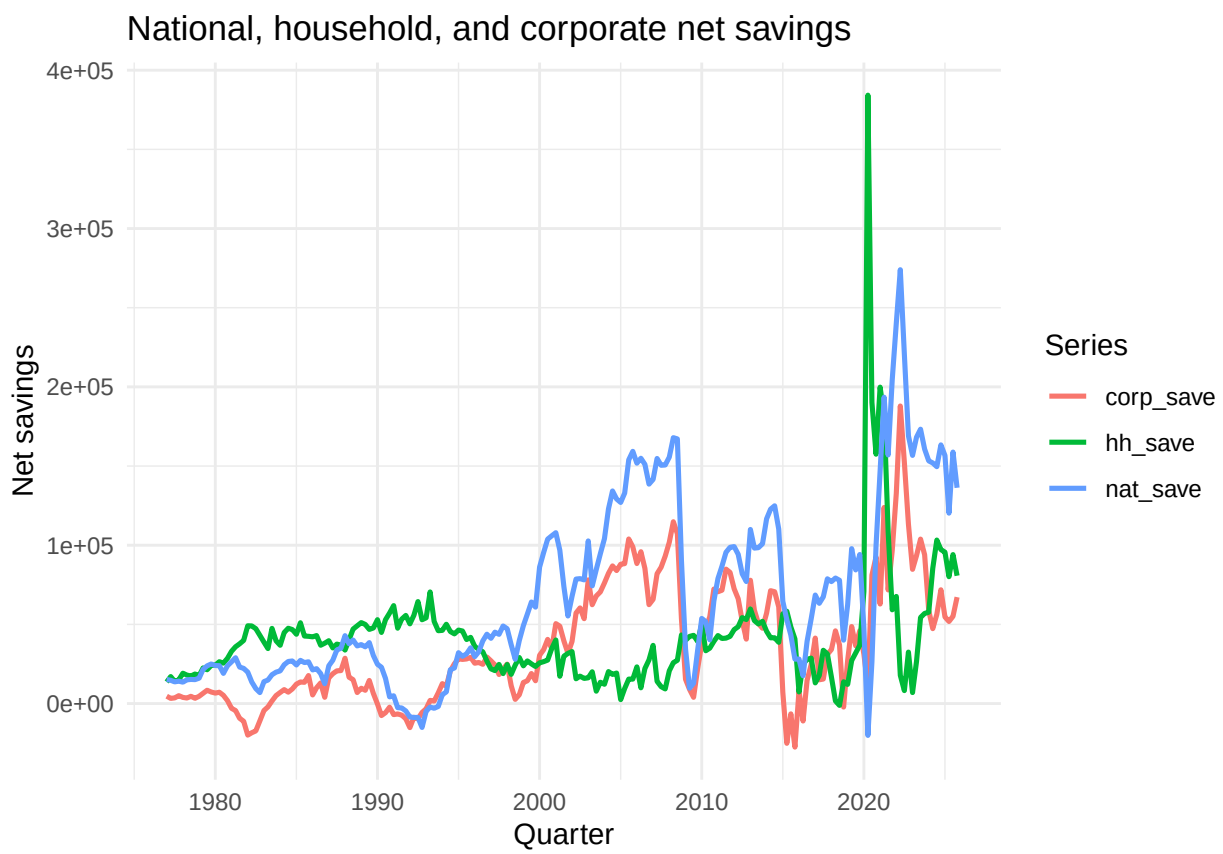
```
## Rows: 197
## Columns: 9
## $ date      <date> 1977-01-01, 1977-04-01, 1977-07-01, 1977-10-01, 1978-01-01,~
## $ nat_save  <dbl> 15180, 14684, 13724, 14136, 13512, 14992, 15232, 15124, 1603~
## $ hh_save   <dbl> 13632, 17048, 13920, 15024, 19168, 17984, 17416, 18660, 1766~
## $ corp_save <dbl> 4660, 3260, 3592, 4984, 3892, 3504, 4576, 3336, 4588, 6492, ~
## $ gdi       <dbl> 37.1, 37.3, 37.4, 37.8, 38.0, 38.3, 38.6, 38.9, 39.3, 40.3, ~
## $ cons      <dbl> 418612, 417567, 424081, 428064, 431165, 434052, 436086, 4373~
## $ unemp     <dbl> 7.8, 7.8, 8.3, 8.5, 8.5, 8.4, 8.4, 8.3, 7.9, 7.4, 7.0, 7.2, ~
## $ cpi       <dbl> 32.8, 33.5, 34.2, 34.9, 35.7, 36.6, 37.1, 37.9, 39.0, 39.9, ~
## $ rate      <dbl> 8.164062, 7.700000, 7.500000, 7.500000, 7.630769, 8.492308, ~
```

```r
summary(eda_df)
```

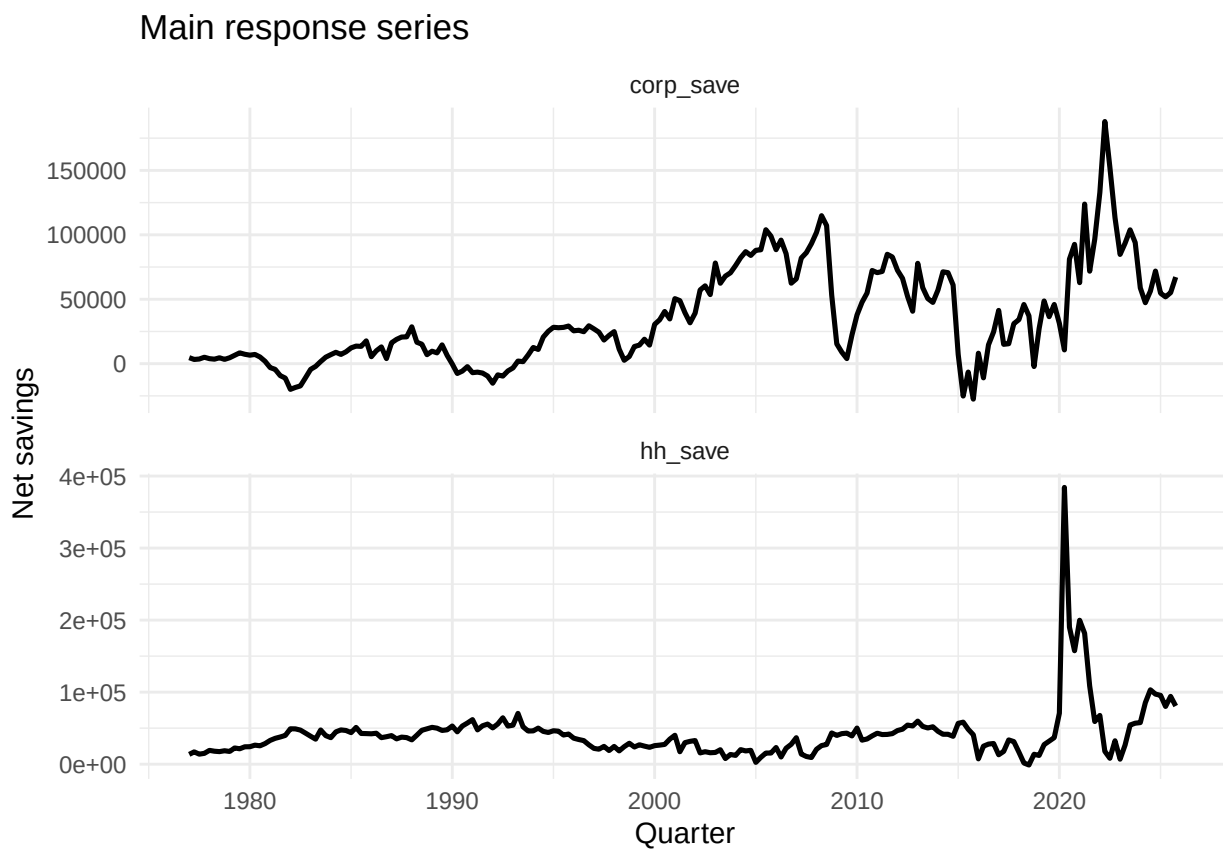
```
##       date               nat_save         hh_save         corp_save
##  Min.   :1977-01-01   Min.   :-20116   Min.   : -1168   Min.   :-27516
##  1st Qu.:1989-04-01   1st Qu.: 23627   1st Qu.: 23350   1st Qu.:  6463
##  Median :2001-07-01   Median : 44448   Median : 37636   Median : 25036
##  Mean   :2001-07-01   Mean   : 66616   Mean   : 41686   Mean   : 35195
##  3rd Qu.:2013-10-01   3rd Qu.: 99984   3rd Qu.: 48448   3rd Qu.: 61384
##  Max.   :2026-01-01   Max.   :274012   Max.   :384240   Max.   :187928
##                       NA's   :1        NA's   :1        NA's   :1
##       gdi              cons             unemp             cpi
##  Min.   : 37.10   Min.   : 417567   Min.   : 4.900   Min.   : 32.80
##  1st Qu.: 51.38   1st Qu.: 570258   1st Qu.: 6.900   1st Qu.: 74.90
##  Median : 71.10   Median : 761540   Median : 7.600   Median : 98.60
##  Mean   : 72.68   Mean   : 826977   Mean   : 8.037   Mean   : 98.83
##  3rd Qu.: 94.42   3rd Qu.:1087645   3rd Qu.: 9.100   3rd Qu.:123.30
##  Max.   :118.50   Max.   :1431258   Max.   :13.100   Max.   :165.00
##  NA's   :1        NA's   :1
##       rate
##  Min.   : 0.500
##  1st Qu.: 1.936
##  Median : 4.674
##  Mean   : 5.532
##  3rd Qu.: 8.569
##  Max.   :20.153
##
```

```r
savings_long <- eda_df |>
  select(date, nat_save, hh_save, corp_save) |>
  pivot_longer(
    cols = -date,
    names_to = "series",
    values_to = "value"
  )

ggplot(savings_long, aes(x = date, y = value, color = series)) +
  geom_line(linewidth = 0.9) +
  theme_minimal() +
  labs(
    title = "National, household, and corporate net savings",
    x = "Quarter",
    y = "Net savings",
    color = "Series"
```

```
  )
```




```
main_response_long <- eda_df |>
  select(date, hh_save, corp_save) |>
  pivot_longer(
    cols = -date,
    names_to = "series",
    values_to = "value"
  )

ggplot(main_response_long, aes(x = date, y = value)) +
  geom_line(linewidth = 0.9) +
  facet_wrap(~ series, scales = "free_y", ncol = 1) +
  theme_minimal() +
  labs(
    title = "Main response series",
    x = "Quarter",
    y = "Net savings"
  )
```

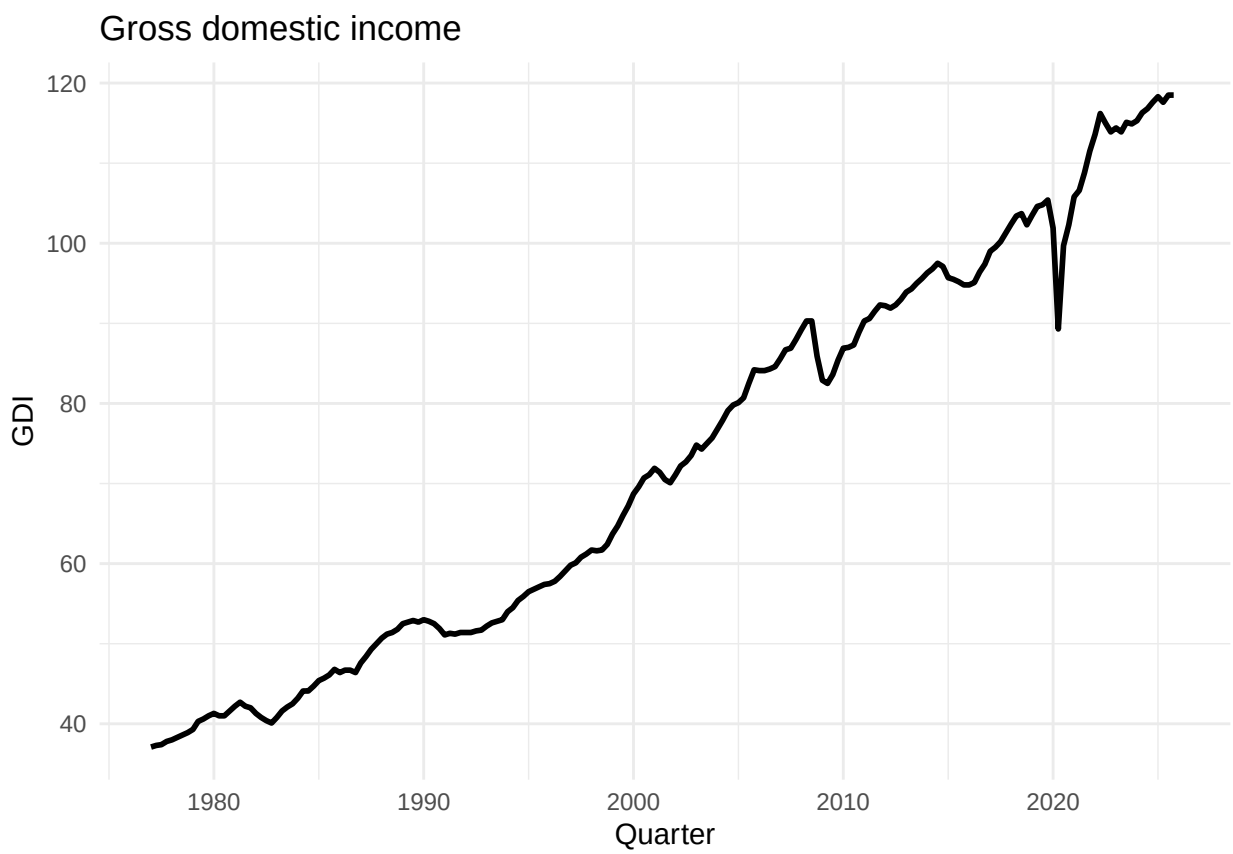



```r
eda_df <- eda_df |>
  mutate(
    gdi_qoq = 100 * (gdi / lag(gdi) - 1),
    gdi_yoy = 100 * (gdi / lag(gdi, 4) - 1),
    lgdi = log(gdi),
    lgdi_yoy = 100 * (lgdi - lag(lgdi, 4))
  )

summary(select(eda_df, gdi, gdi_qoq, gdi_yoy, lgdi_yoy))
```

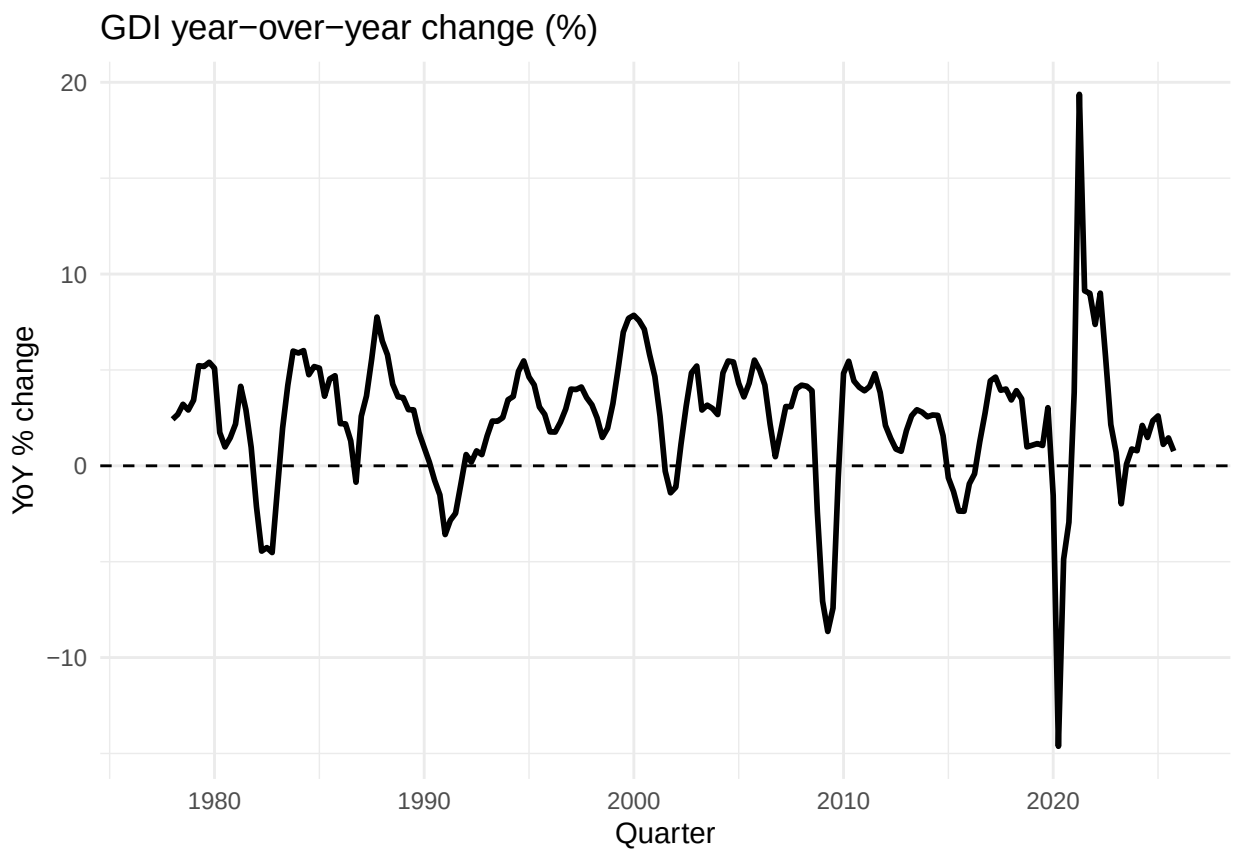
```
##       gdi            gdi_qoq            gdi_yoy           lgdi_yoy
##  Min.   : 37.10   Min.   :-12.3651   Min.   :-14.627   Min.   :-15.814
##  1st Qu.: 51.38   1st Qu.:  0.1387   1st Qu.:  1.042   1st Qu.:  1.037
##  Median : 71.10   Median :  0.7322   Median :  2.852   Median :  2.812
##  Mean   : 72.68   Mean   :  0.6105   Mean   :  2.484   Mean   :  2.398
##  3rd Qu.: 94.42   3rd Qu.:  1.2754   3rd Qu.:  4.269   3rd Qu.:  4.180
##  Max.   :118.50   Max.   : 11.6461   Max.   : 19.373   Max.   : 17.708
##  NA's   :1        NA's   :2          NA's   :5         NA's   :5
```

```r
p1 <- ggplot(eda_df, aes(x = date, y = gdi)) +
  geom_line(linewidth = 1) +
  theme_minimal() +
  labs(
    title = "Gross domestic income",
    x = "Quarter",
    y = "GDI"
  )
```

```r
p2 <- ggplot(eda_df, aes(x = date, y = gdi_yoy)) +
  geom_line(linewidth = 1) +
  geom_hline(yintercept = 0, linetype = "dashed") +
  theme_minimal() +
  labs(
    title = "GDI year-over-year change (%)",
    x = "Quarter",
    y = "YoY % change"
  )

p1
```




```r
p2
```

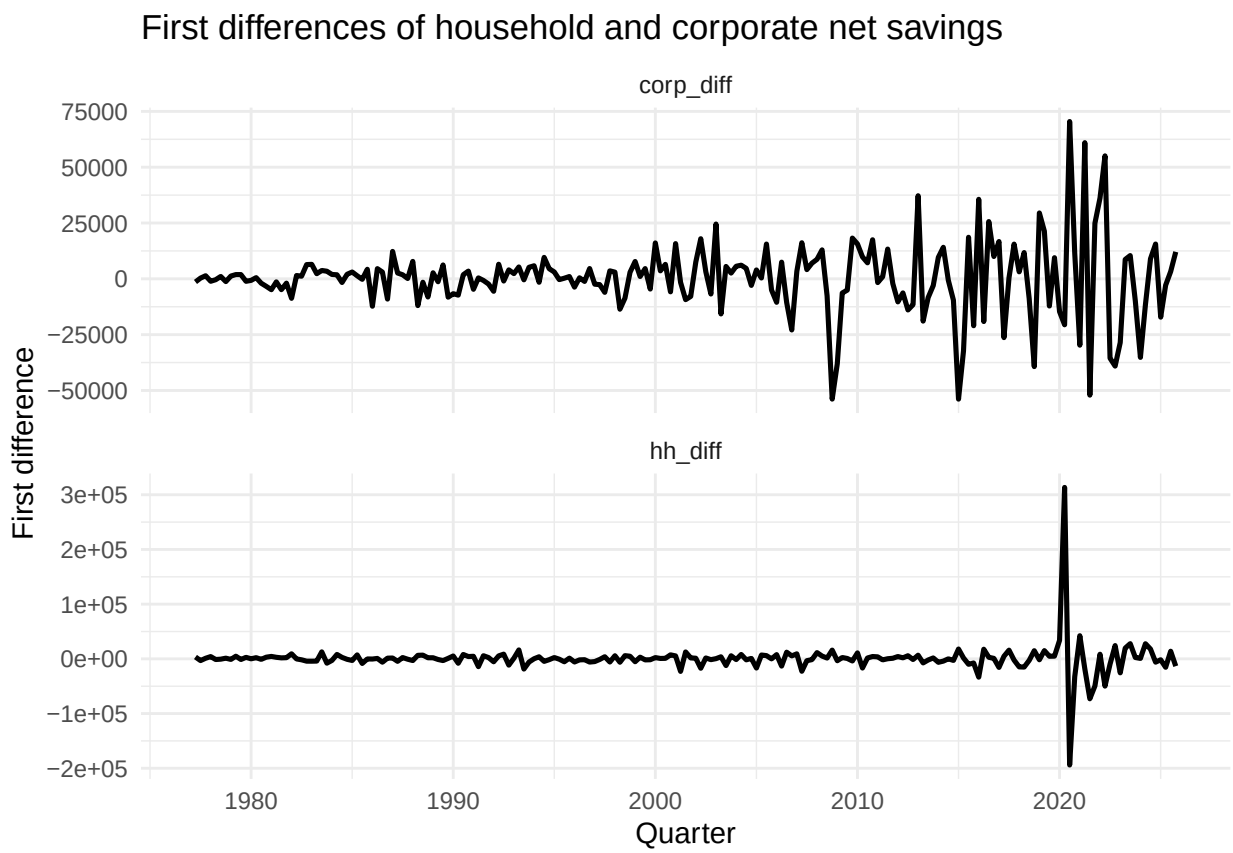



```r
eda_df <- eda_df |>
  mutate(
    hh_diff = hh_save - lag(hh_save),
    corp_diff = corp_save - lag(corp_save),
    nat_diff = nat_save - lag(nat_save),
    unemp_diff = unemp - lag(unemp),
    rate_diff = rate - lag(rate),
    cons_yoy = 100 * (cons / lag(cons, 4) - 1),
    cpi_yoy = 100 * (cpi / lag(cpi, 4) - 1)
  )

summary(select(
  eda_df,
  hh_diff,
  corp_diff,
  nat_diff,
  unemp_diff,
  rate_diff,
  cons_yoy,
  cpi_yoy
))
```

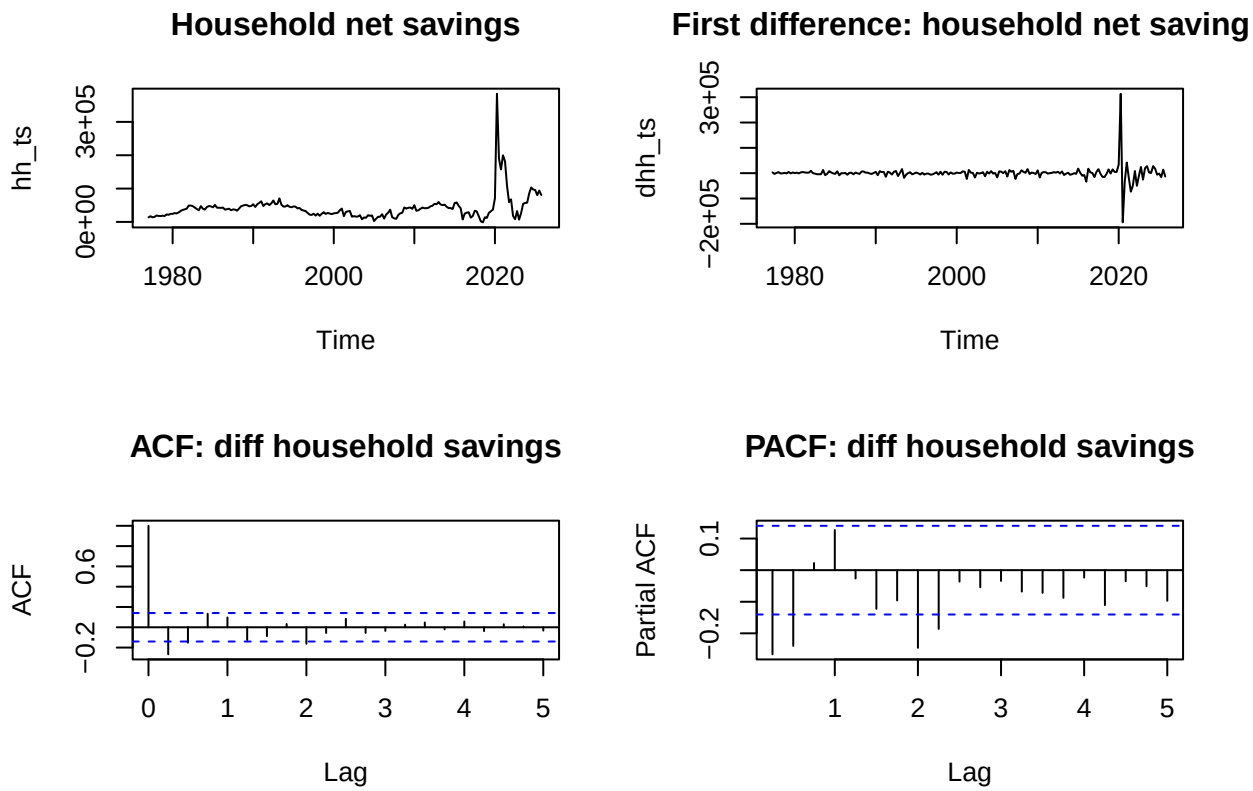
```
##     hh_diff            corp_diff           nat_diff        unemp_diff
##  Min.   :-194076.0   Min.   :-53880.0   Min.   :-76836   Min.   :-3.500000
##  1st Qu.:  -3664.0   1st Qu.: -5708.0   1st Qu.: -4871   1st Qu.:-0.200000
##  Median :    848.0   Median :  1040.0   Median :   792   Median :-0.100000
```

```
##  Mean   :    344.1   Mean   :   320.8   Mean   :   621   Mean   :-0.006633
##  3rd Qu.:   4780.0   3rd Qu.:  6298.0   3rd Qu.:  7130   3rd Qu.: 0.200000
##  Max.   : 313472.0   Max.   : 70464.0   Max.   : 73236   Max.   : 4.100000
##  NA's   :2           NA's   :2          NA's   :2        NA's   :1
##    rate_diff          cons_yoy          cpi_yoy
##  Min.   :-3.6200   Min.   :-15.230   Min.   :-0.8643
##  1st Qu.:-0.3922   1st Qu.:  1.892   1st Qu.: 1.5460
##  Median : 0.0000   Median :  2.786   Median : 2.3519
##  Mean   :-0.0289   Mean   :  2.597   Mean   : 3.3734
##  3rd Qu.: 0.2358   3rd Qu.:  3.913   3rd Qu.: 4.2467
##  Max.   : 3.4486   Max.   : 13.905   Max.   :12.7563
##  NA's   :1         NA's   :5         NA's   :4
```

```r
transformed_long <- eda_df |>
  select(date, hh_diff, corp_diff) |>
  pivot_longer(
    cols = -date,
    names_to = "series",
    values_to = "value"
  )

ggplot(transformed_long, aes(x = date, y = value)) +
  geom_line(linewidth = 0.9) +
  facet_wrap(~ series, scales = "free_y", ncol = 1) +
  theme_minimal() +
  labs(
    title = "First differences of household and corporate net savings",
    x = "Quarter",
    y = "First difference"
  )
```



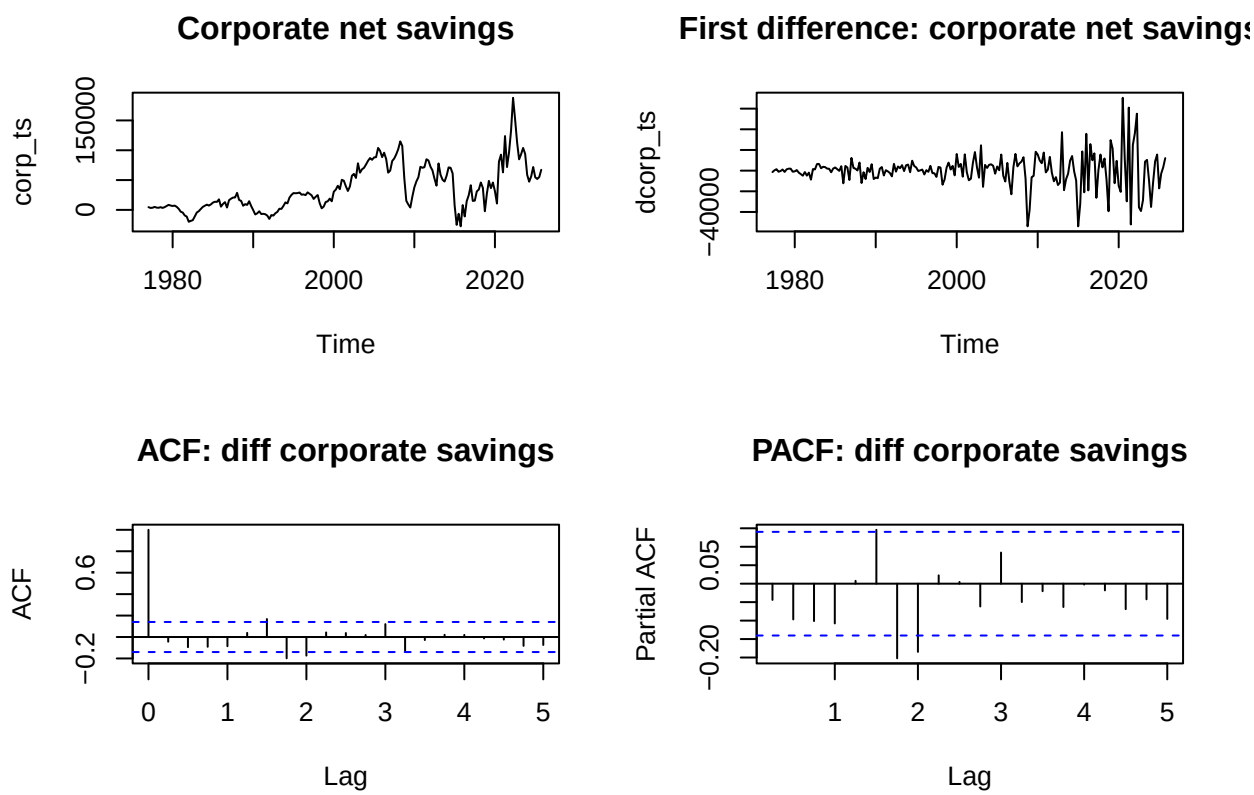

```
hh_ts <- ts(
  eda_df$hh_save,
  start = c(year(min(eda_df$date)), quarter(min(eda_df$date))),
  frequency = 4
)

dhh_ts <- diff(hh_ts)

par(mfrow = c(2, 2))
plot(hh_ts, main = "Household net savings")
plot(dhh_ts, main = "First difference: household net savings")
acf(na.omit(dhh_ts), lag.max = 20, main = "ACF: diff household savings")
pacf(na.omit(dhh_ts), lag.max = 20, main = "PACF: diff household savings")
```




```r
par(mfrow = c(1, 1))

adf.test(na.omit(dhh_ts))
```

```
##
##  Augmented Dickey-Fuller Test
##
## data:  na.omit(dhh_ts)
## Dickey-Fuller = -6.3739, Lag order = 5, p-value = 0.01
## alternative hypothesis: stationary
```

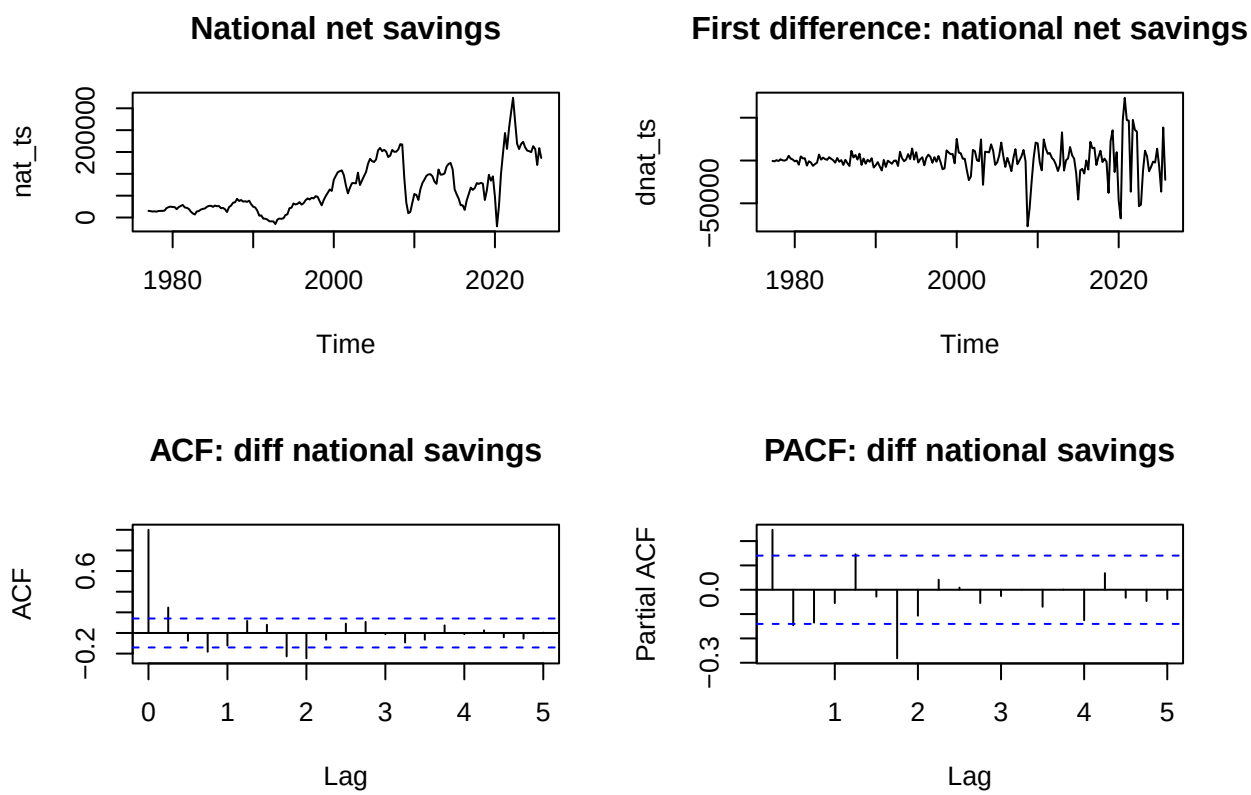
```r
corp_ts <- ts(
  eda_df$corp_save,
  start = c(year(min(eda_df$date)), quarter(min(eda_df$date))),
  frequency = 4
)

dcorp_ts <- diff(corp_ts)

par(mfrow = c(2, 2))
plot(corp_ts, main = "Corporate net savings")
plot(dcorp_ts, main = "First difference: corporate net savings")
acf(na.omit(dcorp_ts), lag.max = 20, main = "ACF: diff corporate savings")
pacf(na.omit(dcorp_ts), lag.max = 20, main = "PACF: diff corporate savings")
```




```r
par(mfrow = c(1, 1))

adf.test(na.omit(dcorp_ts))
```

```
##
##  Augmented Dickey-Fuller Test
##
## data:  na.omit(dcorp_ts)
## Dickey-Fuller = -5.42, Lag order = 5, p-value = 0.01
## alternative hypothesis: stationary
```

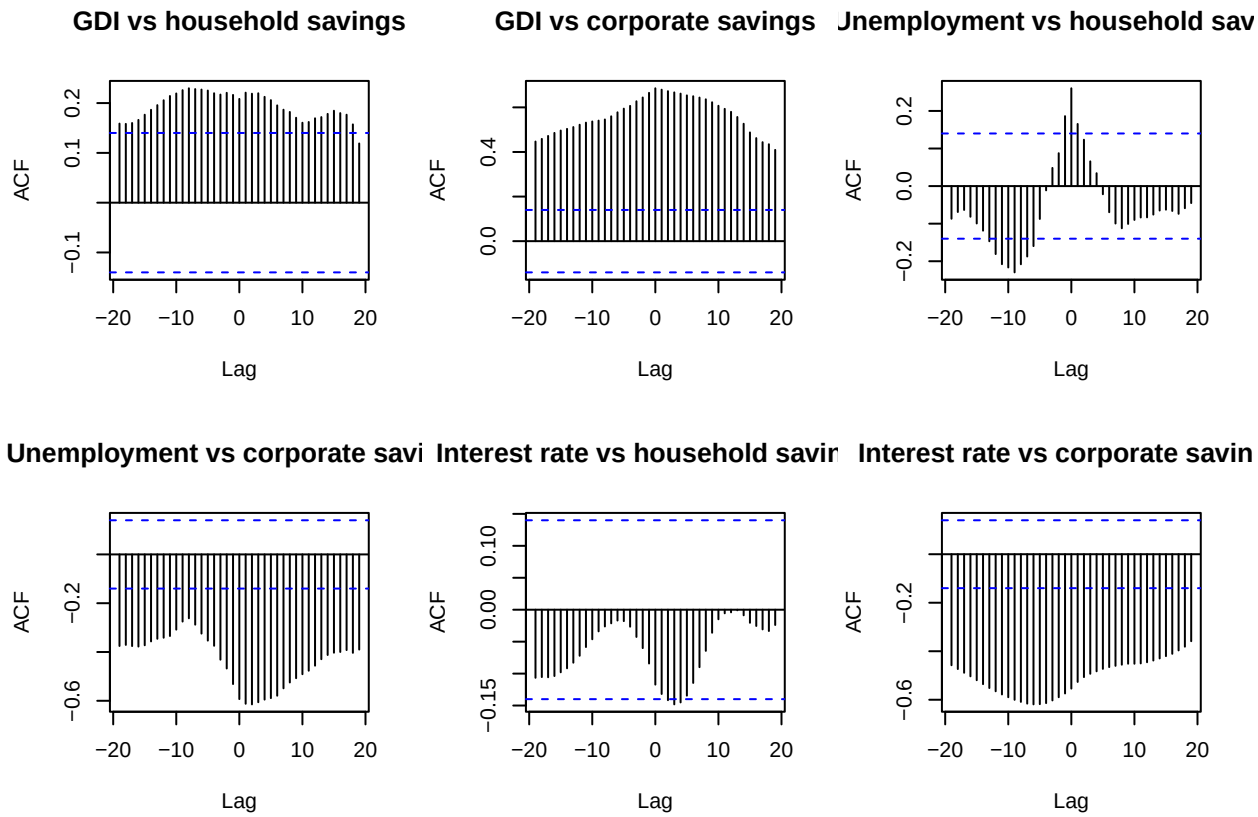
```r
nat_ts <- ts(
  eda_df$nat_save,
  start = c(year(min(eda_df$date)), quarter(min(eda_df$date))),
  frequency = 4
)

dnat_ts <- diff(nat_ts)

par(mfrow = c(2, 2))
plot(nat_ts, main = "National net savings")
plot(dnat_ts, main = "First difference: national net savings")
acf(na.omit(dnat_ts), lag.max = 20, main = "ACF: diff national savings")
pacf(na.omit(dnat_ts), lag.max = 20, main = "PACF: diff national savings")
```



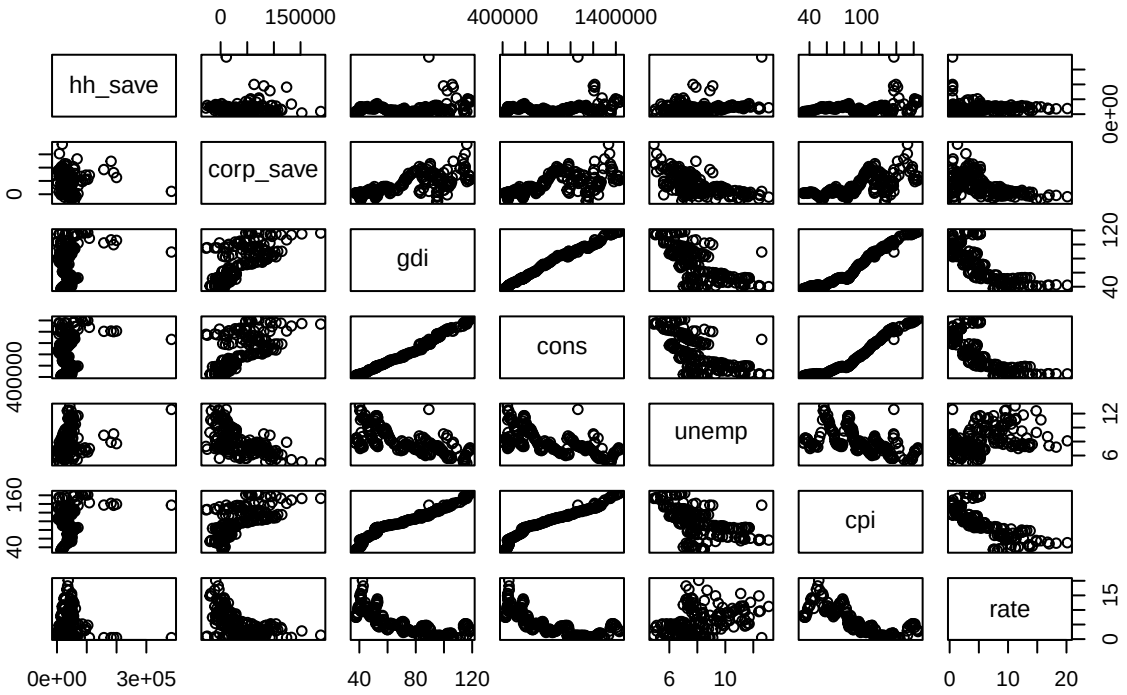

```r
par(mfrow = c(1, 1))

adf.test(na.omit(dnat_ts))
```

```
##
##  Augmented Dickey-Fuller Test
##
## data:  na.omit(dnat_ts)
## Dickey-Fuller = -5.5206, Lag order = 5, p-value = 0.01
## alternative hypothesis: stationary
```

```r
par(mfrow = c(2, 3))

ccf(na.omit(eda_df$gdi), na.omit(eda_df$hh_save),
    main = "GDI vs household savings")
ccf(na.omit(eda_df$gdi), na.omit(eda_df$corp_save),
    main = "GDI vs corporate savings")

ccf(na.omit(eda_df$unemp), na.omit(eda_df$hh_save),
    main = "Unemployment vs household savings")
ccf(na.omit(eda_df$unemp), na.omit(eda_df$corp_save),
    main = "Unemployment vs corporate savings")

ccf(na.omit(eda_df$rate), na.omit(eda_df$hh_save),
    main = "Interest rate vs household savings")
ccf(na.omit(eda_df$rate), na.omit(eda_df$corp_save),
    main = "Interest rate vs corporate savings")
```




```r
par(mfrow = c(1, 1))
```

```r
pairs(
  eda_df |>
    select(hh_save, corp_save, gdi, cons, unemp, cpi, rate) |>
    na.omit(),
  main = "Scatterplot matrix for key variables"
)
```

Scatterplot matrix for key variables